

Counting Problems

Math 252

1. A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?
2. The chairs of an auditorium are labeled with a letter (indicating the row) and a positive integer not exceeding 100 (indicating the seat within the row). Give an upper bound on the number of seats in the auditorium.
3. There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there?
4. How many bitstrings of length 8 are there?
5. How many bitstrings of length 8 either start with a 1 or end with a 1?
6. A state issues license plates that consist of three letters followed by three numbers. How many such plates are there?
7. A graph isomorphism is a 1-1 and onto function from the vertices in one graph to the vertices in another (that additionally preserves edges). You decide to write a little program that will check each 1-1 and onto function to see if it an isomorphism.

If your graphs have n vertices, how many such functions must you check?

8. Now you modify your graph isomorphism algorithm so that it only tries bijections (1-1 and onto functions) that respect the degrees of the vertices. If the degree sequences of two graphs are both $(5, 5, 4, 4, 4, 3, 3, 2)$, how many functions must you check? How much better is this than your previous algorithm (on these two graphs)?
9. Let's use the following coding scheme to describe possible telephone numbers.
 - let X denote a digit that can take on any integer value $0 - 9$;
 - let N denote a digit that can take on values $2 - 9$
 - let Y denote a digit that can only take on values 0 or 1.
 - a. In the scheme used in the 1960's, telephone numbers had to have the form NYX-NNX-XXXX. How many such phone numbers are there?
 - b. A new scheme (*The North American Numbering Plan*) was introduced that required telephone numbers have the form NXX-NXX-XXXX. How many more phone numbers are available in the new scheme?
10. How many password of length 6, 7, or 8 are there if the passwords must be alpha-numeric, capitalization doesn't matter, and there must be at least one digit?
11. In the previous example, how many passwords are there if the first character must be a letter?
12. Professor Pruim puts the names of 30 students into a hat. He draws one out and gives that person a chocolate bar. Then he draws out another name, and gives that person a bag of M and M's. Finally, he draws out a third name and gives that person a bag of Valentine hearts he picked up cheap on February 15.
 - a. How many different ways can the prizes be distributed?
 - b. If you are one of the 30 students, how many of those ways include you getting a prize?

13. Below is a description of IPv4 addresses.

Counting Internet Addresses In the Internet, which is made up of interconnected physical networks of computers, each computer (or more precisely, each network connection of a computer) is assigned an *Internet address*. In Version 4 of the Internet Protocol (IPv4), now in use, an address is a string of 32 bits. It begins with a *network number (netid)*. The netid is followed by a *host number (hostid)*, which identifies a computer as a member of a particular network.

Three forms of addresses are used, with different numbers of bits used for netids and hostids. **Class A addresses**, used for the largest networks, consist of 0, followed by a 7-bit netid and a 24-bit hostid. **Class B addresses**, used for medium-sized networks, consist of 10, followed by a 14-bit netid and a 16-bit hostid. **Class C addresses**, used for the smallest networks, consist of 110, followed by a 21-bit netid and an 8-bit hostid. There are several restrictions on addresses because of special uses: 1111111 is not available as the netid of a Class A network, and the hostids consisting of all 0s and all 1s are not available for use in any network. A computer on the Internet has either a Class A, a Class B, or a Class C address. (Besides Class A, B, and C addresses, there are also Class D addresses, reserved for use in multicasting when multiple computers are addressed at a single time, consisting of 1110 followed by 28 bits, and Class E addresses, reserved for future use, consisting of 11110 followed by 27 bits. Neither Class D nor Class E addresses are assigned as the IP address of a computer on the Internet.) Figure 1 illustrates IPv4 addressing. (Limitations on the number of Class A and Class B netids have made IPv4 addressing inadequate; IPv6, a new version of IP, uses 128-bit addresses to solve this problem.)

How many different IPv4 addresses are available for computers on the Internet?

Bit Number	0	1	2	3	4	8	16	24	31	
Class A	0	netid				hostid				
Class B	1	0	netid				hostid			
Class C	1	1	0	netid				hostid		
Class D	1	1	1	0	Multicast Address					
Class E	1	1	1	1	0	Address				

Answers/Solutions

- $12 \cdot 11 = 132$.
- $26 \cdot 100 = 2600$
- $32 \cdot 24 = 768$.
- $2^8 = 256$
- Two ways:
 - only bad thing is beginning AND ending with 0, so $2^8 - 2^6 = 192$.
 - Start with 1 or end with 1 but not both: $2^7 + 2^7 - 2^6 = 192$
- $10^3 \cdot 26^3 = 17576000$
- $n!$
- $2 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 = 24$. ($8! = 40320$, so that's a pretty big speed up.)

9. See book.
10. $36^6 - 26^6 + 36^7 - 26^7 + 36^8 - 26^8 = 2684483063360$
11. $26 * (36^5 - 26^5) + 26 * (36^6 - 26^6) + 26 * (36^7 - 26^7) = 1878468937280$
12. a. $30 * 29 * 28 = 24360$
 b. $1 * 29 * 28 + 29 * 1 * 28 + 29 * 28 * 1 = 30 * 29 * 28 - 29 * 28 * 27 = 2436$
13. See book.